Causal Assumptions

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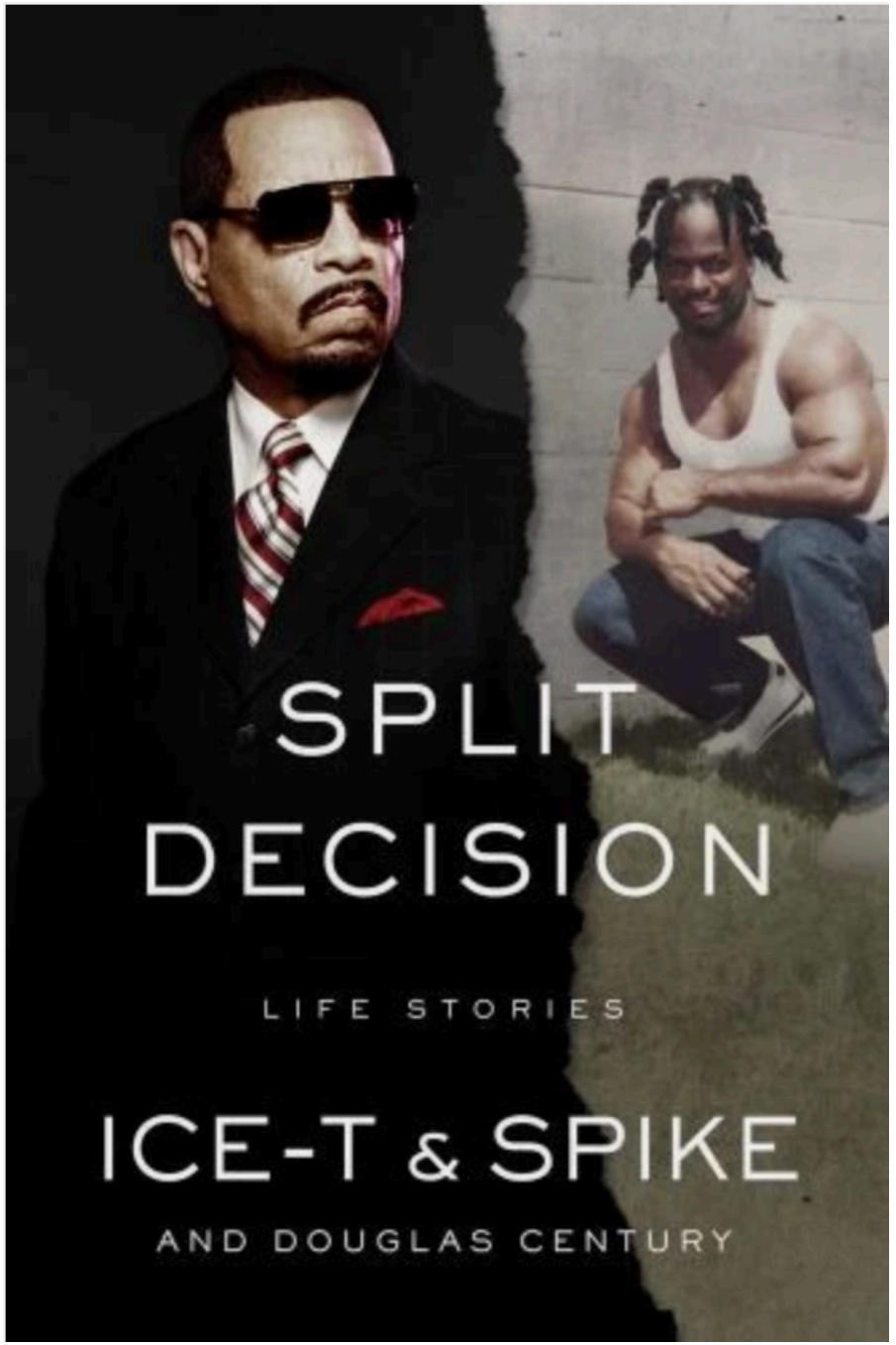

SPLIT
DECISION
LIFE STORIES
ICE-T & SPIKE
AND DOUGLAS CENTURY

Are they good proxies for each other?

Person	Observed	Missing counterfactual
Ice-T	Abandons criminal life: fame and fortune	Does one more heist: 35 years in prison
Spike	Does one more heist: 35 years in prison	Abandons criminal life: fame and fortune

They differ in more than their choices

- Ice-T's decision was not the only factor in his success: he had enough musical talent to make a career of it

- Spike might differ in hard-to-measure ways, which makes him a poorer proxy than he first seems

We must rely on statistical techniques to help construct these unobservable counterfactuals from observed data

Potential outcomes

- Factual and counterfactual outcomes are two realizations of potential outcomes

- Say the exposure has two levels, `x = continue` doing jewel heists or `x = quit`

- There are two potential outcomes, `y(continue)` and `y(quit)`, and only one will be realized

The missing potential outcome

- One potential outcome is observable and one is missing; early causal inference methods were framed as missing data problems

- We're missing `y(continue)` for Ice-T and `y(quit)` for Spike, so we can't calculate their individual causal effects

Causal assumptions allow us to imbue observed data with the meaning of potential outcomes and their analysis with causal effects

Does chocolate make you happier?

Suppose some happiness index exists that ranges from 1 to 10

```r
data <- tibble(
  id = 1:10,
  y_chocolate = c(4, 4, 6, 5, 6, 5, 6, 7, 5, 6),
  y_vanilla = c(1, 3, 4, 5, 5, 6, 8, 6, 3, 5)
) |>
  mutate(causal_effect = y_chocolate - y_vanilla)

data |>
  summarize(
    avg_chocolate = mean(y_chocolate),
    avg_vanilla = mean(y_vanilla),
    avg_causal_effect = mean(causal_effect)
  )
```

```
# A tibble: 1 × 3
  avg_chocolate avg_vanilla avg_causal_effect
          <dbl>       <dbl>             <dbl>
1           5.4         4.6               0.8
```

Both potential outcomes are known

	Potential outcomes		Causal effect
ID	y(chocolate)	y(vanilla)	y(chocolate) - y(vanilla)
1	4	1	3
2	4	3	1
3	6	4	2
4	5	5	0
5	6	5	1
6	5	6	-1
7	6	8	-2
8	7	6	1
9	5	3	2
10	6	5	1

We can calculate every individual causal effect

In reality, each person eats one flavor

```r
set.seed(11)
data_observed <- data |>
  mutate(
    exposure = if_else(
      rbinom(n(), 1, 0.5) == 1,
      "chocolate",
      "vanilla"
    ),
    observed_outcome = case_when(
      exposure == "chocolate" ~ y_chocolate,
      exposure == "vanilla" ~ y_vanilla
    )
  )

data_observed |>
  group_by(exposure) |>
  summarise(avg_outcome = mean(observed_outcome))
```

```
# A tibble: 2 × 2
  exposure  avg_outcome
  <chr>           <dbl>
1 chocolate        5.33
2 vanilla          4.71
```

We're a little off because of the low sample size

Only the observed outcome remains

	Potential outcomes		Causal effect
ID	y(chocolate)	y(vanilla)	y(chocolate) - y(vanilla)
1		1	
2		3	
3	6		
4		5	
5		5	
6	5		
7		8	
8		6	
9	5		
10		5	

We can’t calculate the individual causal effect

The three assumptions

- Unconfoundedness methods assume **exchangeability**, **positivity**, and **consistency**

- Knowing a method's assumptions is essential for using it correctly; these are sometimes called identifiability conditions

Exchangeability

Exposure status is independent of the potential outcomes; that is, each exposure group has the same potential outcomes on average

What if we mixed up the labels?

Say we accidentally gave chocolate to the “vanilla” group and vanilla to the “chocolate” group

```r
mix_up <- function(flavor) {
  if_else(flavor == "chocolate", "vanilla", "chocolate")
}

data_observed <- data_observed |>
  mutate(
    exposure = mix_up(exposure),
    observed_outcome = case_when(
      exposure == "chocolate" ~ y_chocolate,
      exposure == "vanilla" ~ y_vanilla
    )
  )

data_observed |>
  group_by(exposure) |>
  summarise(avg_outcome = mean(observed_outcome))
```

```
# A tibble: 2 × 2
  exposure  avg_outcome
  <chr>           <dbl>
1 chocolate        5.43
2 vanilla          4.33
```

Flavor assignment is independent of the potential outcomes

What if participants chose their own flavor?

```r
set.seed(113)
data_observed_exch <- data |>
  mutate(
    prefer_chocolate = y_chocolate > y_vanilla,
    exposure = case_when(
      # people choose the flavor they prefer 80% of the time
      prefer_chocolate ~ if_else(
        rbinom(n(), 1, 0.8) == 1,
        "chocolate",
        "vanilla"
      ),
      !prefer_chocolate ~ if_else(
        rbinom(n(), 1, 0.8) == 1,
        "vanilla",
        "chocolate"
      )
    ),
    observed_outcome = case_when(
      exposure == "chocolate" ~ y_chocolate,
      exposure == "vanilla" ~ y_vanilla
    )
  )

data_observed_exch |>
  group_by(exposure) |>
  summarise(avg_outcome = mean(observed_outcome))
```

```
# A tibble: 2 × 2
  exposure  avg_outcome
  <chr>           <dbl>
1 chocolate        5.29
2 vanilla          5.67
```

Now it seems like vanilla makes you happier!

The groups are no longer exchangeable

Preference influences both flavor and happiness

Conditional exchangeability

Exchangeability can hold within preference groups

But few values fall in each combination

Positivity

- Everyone has a non-zero probability of each exposure level

- **Stochastic violations** are chance occurrences with no observations for a given exposure level

- **Structural violations** happen when an individual cannot receive an exposure level by definition

A stochastic violation

Participants choose the flavor they prefer 80% of the time, as before

```
data_observed_pos |>
  count(prefer_chocolate, exposure) |>
  complete(
    prefer_chocolate,
    exposure = c("chocolate", "vanilla"),
    fill = list(n = 0)
  )
```

```
# A tibble: 4 × 3
  prefer_chocolate exposure     n
  <lgl>            <chr>    <int>
1 FALSE            chocolate    0
2 FALSE            vanilla      3
3 TRUE             chocolate    7
4 TRUE             vanilla      0
```

Positivity must hold within the covariates needed for exchangeability

A structural violation: an allergy to vanilla

Everyone is randomized to a flavor, as before

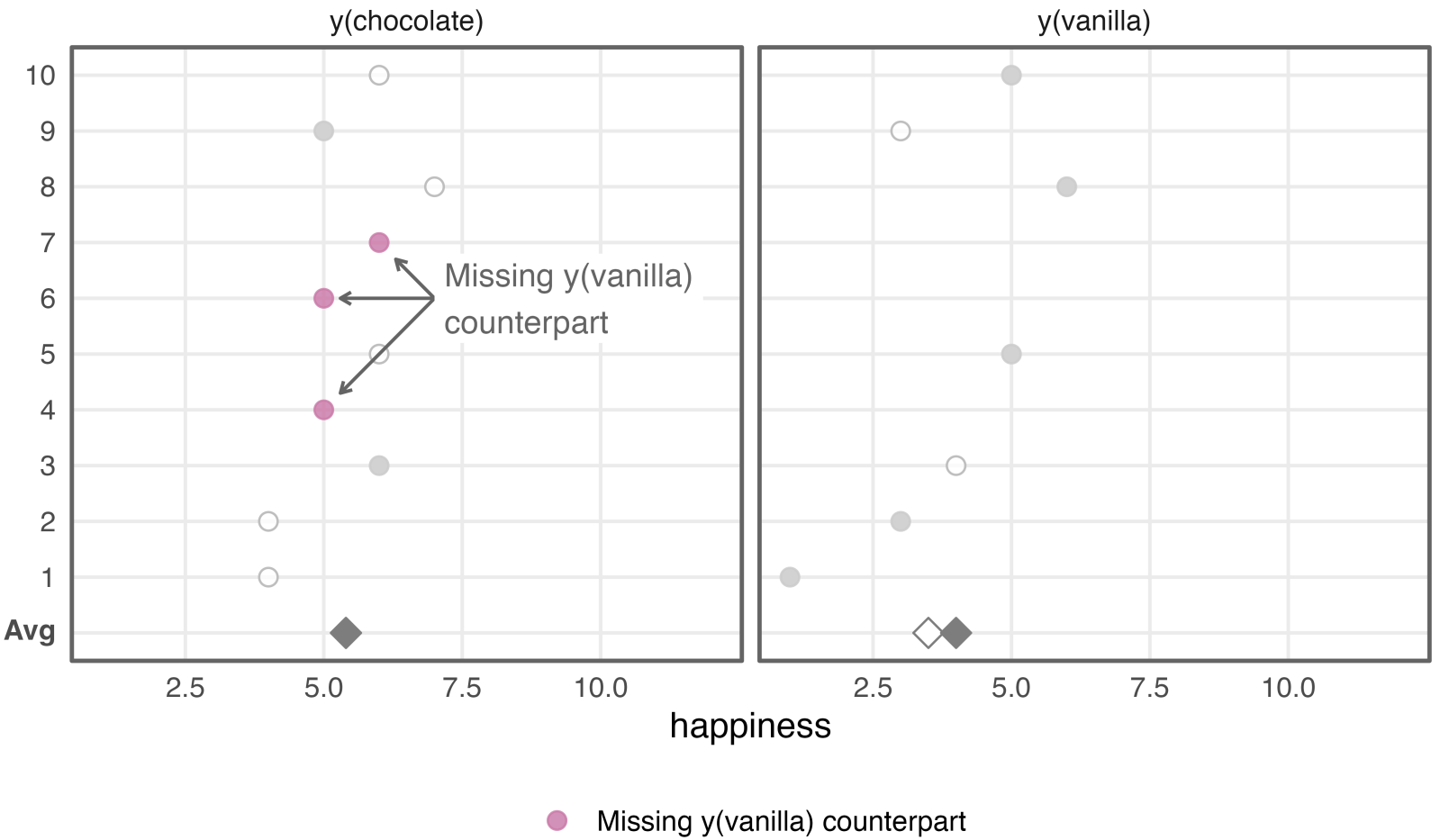
```
set.seed(1)
data_observed_struc <- data_observed_struc |>
  mutate(
    allergy = rbinom(n(), 1, 0.3) == 1,
    # `y_vanilla` is impossible for those with an allergy
    exposure = if_else(allergy, "chocolate", exposure),
    y_vanilla = if_else(allergy, NA, y_vanilla),
    observed_outcome = case_when(
      allergy ~ y_chocolate,
      exposure == "chocolate" ~ y_chocolate,
      exposure == "vanilla" ~ y_vanilla
    )
  )

data_observed_struc |>
  group_by(exposure) |>
  summarise(avg_outcome = mean(observed_outcome))
```

```
# A tibble: 2 × 2
  exposure  avg_outcome
  <chr>           <dbl>
1 chocolate         5.4
2 vanilla           4
```

For those with a vanilla allergy, $y(\texttt{vanilla})$ is not defined

Undefined potential outcomes



We can collect more data, use a statistical model to extrapolate, add eligibility criteria, or change the estimand

Finding positivity violations

- We planted the violations in these examples, so we knew where to look

- In real data, we have to search the covariates for regions where an exposure level is rare or absent

A data set with two violations

- `region` b with `x2` above 1 is never exposed
- Exposure depends steeply on `x1`, so its tails are nearly deterministic

```
1 pos_violations
```

```
# A tibble: 1,000 × 4
   exposure     x1      x2 region
      <int>  <dbl>   <dbl> <fct>
 1        1  0.982 -0.0517 a
 2        1  0.469 -0.499  a
 3        0 -0.108 -0.903  a
 4        0 -0.213 -0.372  a
 5        1  1.16  -0.188  a
 6        1  1.29  -0.737  a
 7        1  0.535 -0.876  b
 8        0 -0.127 -1.40   b
 9        0 -1.22   0.395  b
10        0 -1.12  -0.629  b
# ℹ 990 more rows
```

Positivity regression trees

```
port <- check_port(pos_violations, exposure, c(x1, x2, region))
port
```

```
── PoRT subgroups ──────────────────────────────────────────────────────────────
Exposure: "exposure" (binary)
Observations: 1000
Reading rule: alpha = 0.05, gamma = 2
Prevalence threshold beta: 0.05
Subgroups: 231 reported, 3 with low support
```

Low-support subgroups

```
port |>
  tidy() |>
  filter(low_support) |>
  select(description, exposure_level, n, prevalence)
```

```
# A tibble: 3 × 4
  description           exposure_level     n prevalence
  <chr>                 <chr>          <int>      <dbl>
1 x1<-0.5027            1                305     0.0459
2 x1>=0.9626 & x1<1.154 1                 53     0.962
3 x2>=1.063 & region=b  1                 69     0
```

Structural or stochastic?

- The tree locates the region; only background knowledge decides which kind of violation it is
- The search sees the covariates you give it, so it cannot find a violation defined by a variable you left out

Consistency

- The causal question you claim you are answering is consistent with the one you are actually answering

- The potential outcome of a given treatment value equals the value we actually observe when someone is assigned that treatment value

It seems almost silly when you say it. What else would it be?

Two ways consistency fails

- **Poorly-defined exposure**: multiple treatment versions exist, from surgeries to income to education

- **Interference**: the outcome for any subject depends on another subject's exposure

Two containers of chocolate, one spoiled

Everyone has a third potential outcome, `y_spoiled_chocolate`, which is 0

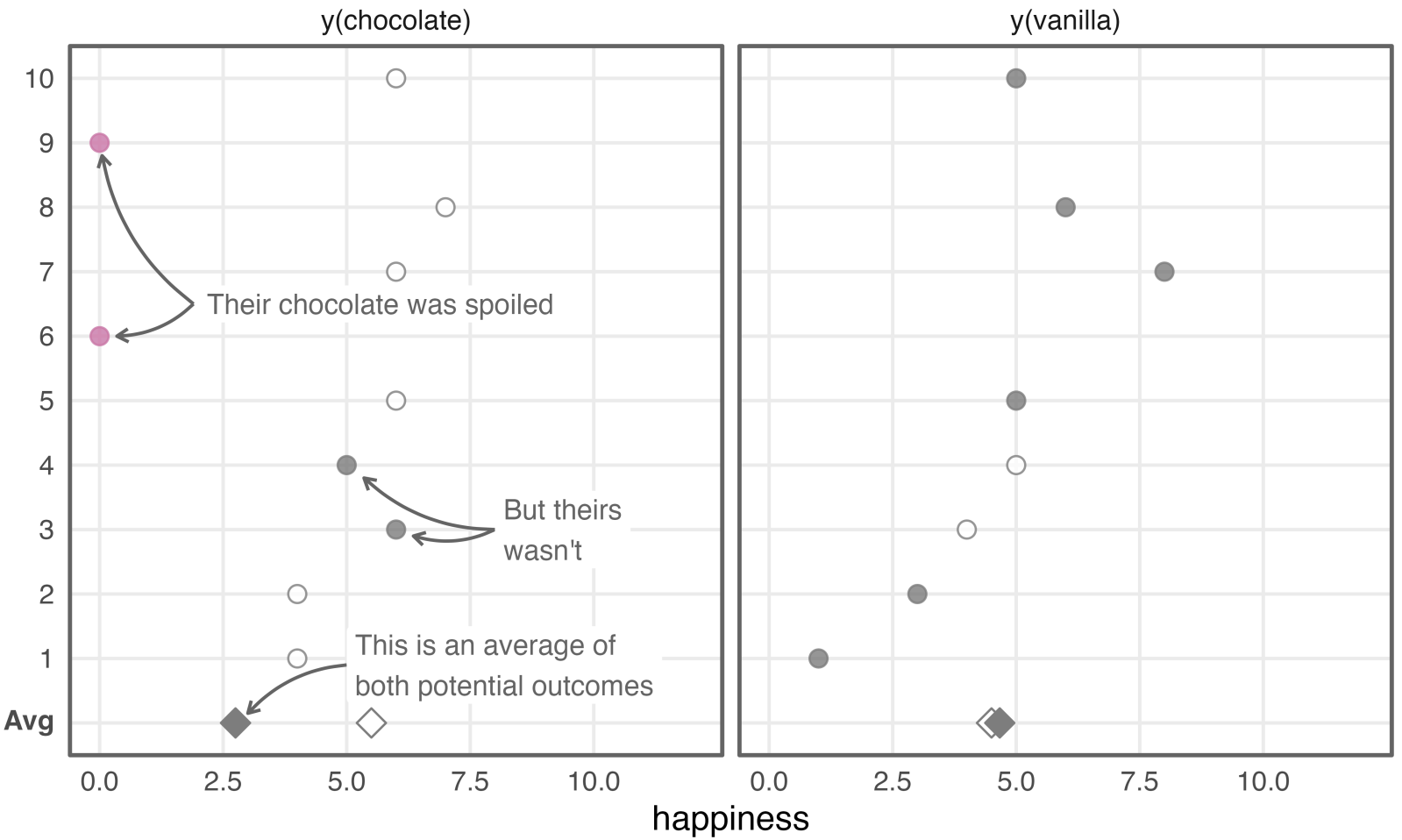
```
set.seed(11)
data_observed_poorly_defined <- data |>
  mutate(
    exposure_unobserved = case_when(
      rbinom(n(), 1, 0.25) == 1 ~ "chocolate (spoiled)",
      rbinom(n(), 1, 0.25) == 1 ~ "chocolate",
      .default = "vanilla"
    ),
    observed_outcome = case_when(
      exposure_unobserved == "chocolate (spoiled)" ~ y_spoiled_chocolate,
      exposure_unobserved == "chocolate" ~ y_chocolate,
      exposure_unobserved == "vanilla" ~ y_vanilla
    ),
    # both containers are recorded as "chocolate"
    exposure = if_else(exposure_unobserved == "vanilla", "vanilla", "chocolate")
  )

data_observed_poorly_defined |>
  group_by(exposure) |>
  summarise(avg_outcome = mean(observed_outcome))
```

```
# A tibble: 2 × 2
  exposure  avg_outcome
  <chr>           <dbl>
1 chocolate        2.75
2 vanilla          4.67
```

The true causal effect of unspoiled chocolate is 0.8, but our estimate is about -1.9

The chocolate group is a mixture



We're treating fresh and spoiled chocolate as the same exposure; the same goes for brands, timing, and portion size

Separate the potential outcomes

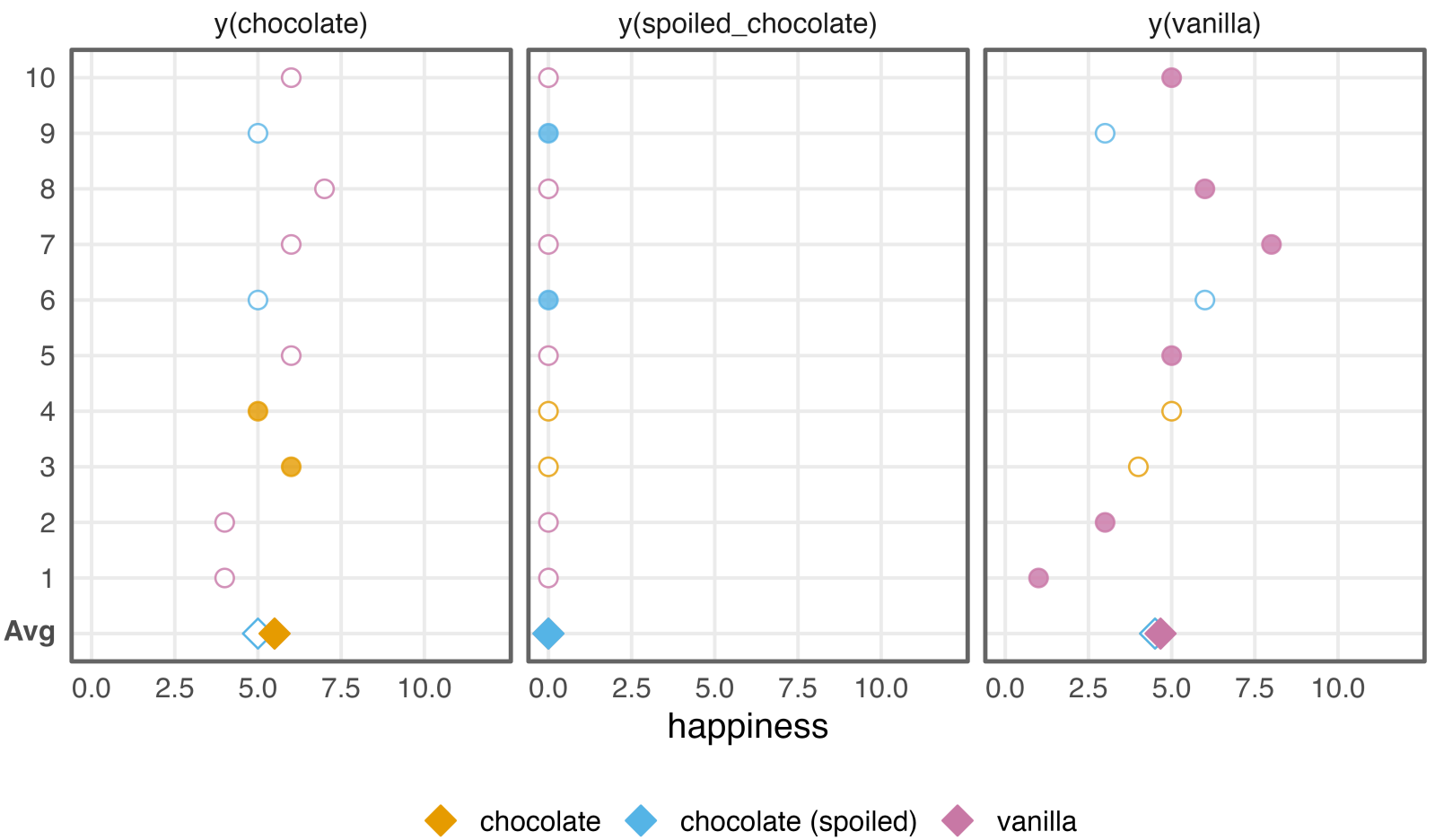
Say we tested the containers after the experiment and know which ones were spoiled

```
data_observed_poorly_defined |>
  group_by(exposure_unobserved) |>
  summarise(avg_outcome = mean(observed_outcome))
```

```
# A tibble: 3 × 2
  exposure_unobserved avg_outcome
  <chr>                     <dbl>
1 chocolate                  5.5
2 chocolate (spoiled)        0
3 vanilla                    4.67
```

Now we're back to the right answer because we've correctly separated the potential outcomes

The data match potential outcomes



There will almost always be some consistency violation; the question is whether it is meaningful

Interference

- An individual's exposure impacts another's potential outcome, as when someone's vaccination affects someone else's risk

- Here, each person has a partner, and having a partner with a different flavor increases happiness by two units

Each person has a partner

```r
data <- tibble(
  id = 1:10,
  partner_id = c(1, 1, 2, 2, 3, 3, 4, 4, 5, 5),
  y_chocolate_chocolate = c(4, 4, 6, 5, 6, 5, 6, 7, 5, 6),
  y_vanilla_vanilla = c(1, 3, 4, 5, 5, 6, 8, 6, 3, 5)
) |>
  mutate(
    y_chocolate_vanilla = y_chocolate_chocolate + 2,
    y_vanilla_chocolate = y_vanilla_vanilla + 2
  )
```

Randomize each individual and their partner

```r
set.seed(37)
data_observed_interf <- data |>
  mutate(
    exposure = if_else(rbinom(n(), 1, 0.5) == 1, "chocolate", "vanilla"),
    exposure_partner = if_else(
      rbinom(n(), 1, 0.5) == 1,
      "chocolate",
      "vanilla"
    ),
    observed_outcome = case_when(
      exposure == "chocolate" & exposure_partner == "chocolate" ~
        y_chocolate_chocolate,
      exposure == "chocolate" & exposure_partner == "vanilla" ~
        y_chocolate_vanilla,
      exposure == "vanilla" & exposure_partner == "chocolate" ~
        y_vanilla_chocolate,
      exposure == "vanilla" & exposure_partner == "vanilla" ~ y_vanilla_vanilla
    )
  )

data_observed_interf |>
  group_by(exposure) |>
  summarise(avg_outcome = mean(observed_outcome))
```

```
# A tibble: 2 × 2
  exposure  avg_outcome
  <chr>           <dbl>
1 chocolate        6.25
2 vanilla          6.33
```

Each person has three counterfactuals now, and these averages don't estimate any of them

Specify the potential outcomes

```
1 data_observed_interf |>
2   group_by(exposure, exposure_partner) |>
3   summarise(avg_outcome = mean(observed_outcome), .groups = "drop")
```

```
# A tibble: 4 × 3
  exposure  exposure_partner avg_outcome
  <chr>     <chr>                  <dbl>
1 chocolate chocolate               5.5
2 chocolate vanilla                 7
3 vanilla   chocolate               6.75
4 vanilla   vanilla                 5.5
```

One way to combat interference is to change the unit: randomize the partnerships, a cluster randomized trial

Randomize the partnerships

```r
set.seed(11)
partners <- tibble(
  partner_id = 1:5,
  exposure = if_else(rbinom(5, 1, 0.5) == 1, "chocolate", "vanilla")
)

partners_observed <- data |>
  left_join(partners, by = "partner_id") |>
  mutate(
    exposure_partner = exposure,
    observed_outcome = case_when(
      exposure == "chocolate" & exposure_partner == "chocolate" ~
        y_chocolate_chocolate,
      exposure == "vanilla" & exposure_partner == "vanilla" ~ y_vanilla_vanilla
    )
  )

partners_observed |>
  group_by(exposure) |>
  summarise(avg_outcome = mean(observed_outcome))
```

```
# A tibble: 2 × 2
  exposure  avg_outcome
  <chr>           <dbl>
1 chocolate        5.5
2 vanilla          4.38
```

There is no interference between different partner sets

Malaria nets, revisited

Malaria nets and the assumptions

- **Exchangeability**: we saw what can happen with an unmeasured confounder, genetic resistance to malaria

- **Positivity**: a violation would occur if any household would always or never use a bed net, or if no one of low economic status in cold weather ever used one

Consistency and interference

- **Consistency**: a “bed net” can mean many things; does the manufacturer matter? The type of insecticide? The amount?

- Interference is a real problem for bed-net research, though households as the unit reduce some of it

When do study designs support causal inference?

Randomized trials

- In the limit, we expect exchangeability: randomization is the only cause of exposure
- Randomization guarantees positivity so long as no one has a deterministic exposure
- Trials help, but do not guarantee, consistency; interference can still happen

Trials in practice

- In actual trials, exchangeability can fail by chance, people do not adhere to their assigned exposure, and people drop out

- Observational studies have none of a trial's guarantees, even ideal ones

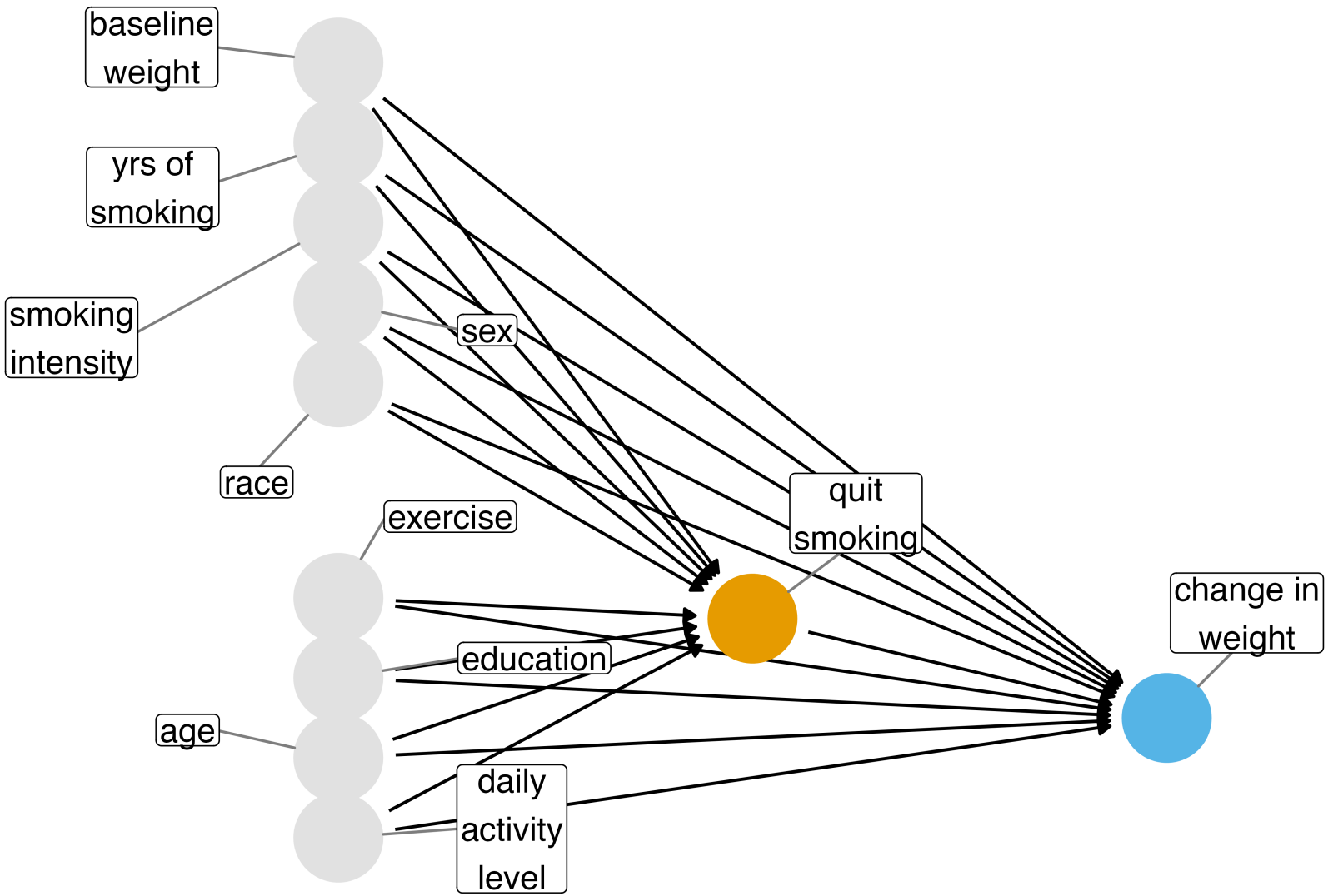
A randomized design is sometimes called an A/B test

Assumptions solved by study design

Assumption	Ideal randomized trial	Realistic randomized trial	Observational study
Consistency (well-defined exposure)	😄	🤷‍♂️	🤷‍♂️
Consistency (no interference)	🤷‍♂️	🤷‍♂️	🤷‍♂️
Positivity	😄	😄	🤷‍♂️
Exchangeability	😄	🤷‍♂️	🤷‍♂️

😄 solved by default, 🤷‍♂️ solvable but not solved by default

Quitting smoking and weight gain



Your Turn

Open `07-causal-assumptions-exercises.qmd`

Discuss with your neighbors: what do exchangeability, positivity, and consistency require for the question, does quitting smoking cause weight gain?

10:00